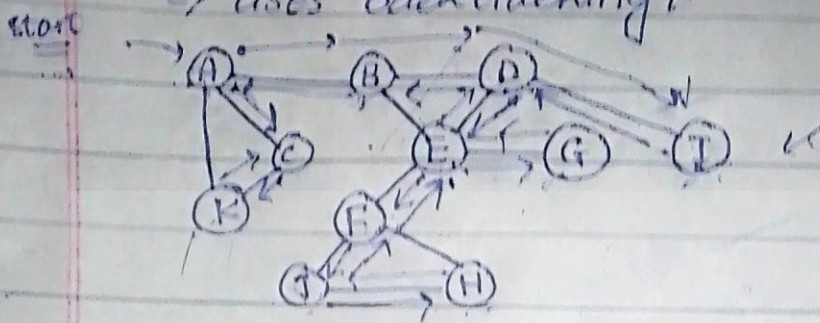


Graph Traversals

ii) DFS (Depth-First Traversal Search)

→ uses LIFO-Queue (stack)

→ uses backtracking



stack: A ✓

output:

stack: A, B ✓

output: ~~A~~

stack: A, B, D ✓

output:

stack: A, B, D, I ✓

output: I ✓

stack: A, B, D, E

output: I

stack: A, B, D, E, F

output: I

stack: A, B, D, E, F, J

output: I

stack: A, B, D, E, F, J, H

output: I, H

stack: A, B, D, E, F, J, H

output: I, H, J

stack: A, B, D, E, F

output: I, H, J, F

stack: A, B, D, E, G

output: I, H, J, F, G

stack: A, B, D, E

output: I, H, J, F, G, E

stack: A, B, D

output: I, H, J, F, G, E, D

stack: A, B

output: I, H, J, F, G, E, D, B

stack: A, C

output: I, H, J, F, G, E, D, B

stack: A, C, K

output: I, H, J, F, G, E, D, B, K

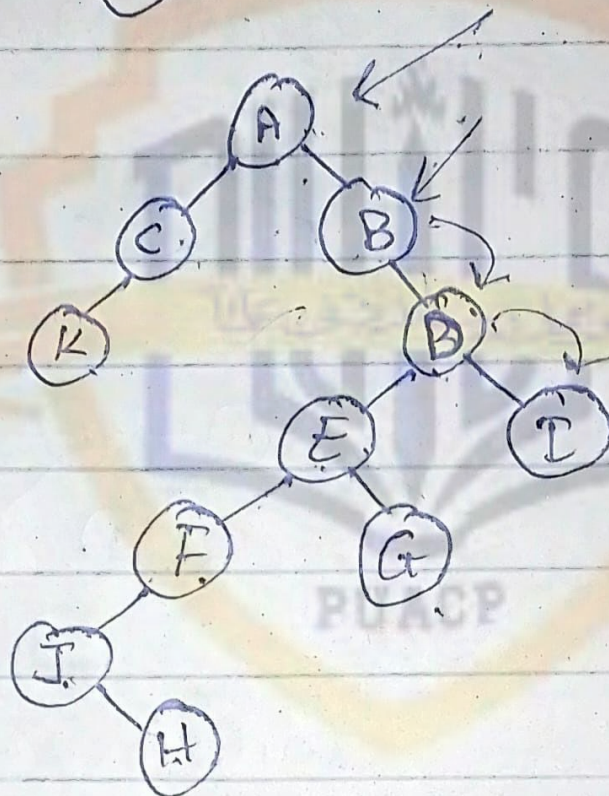
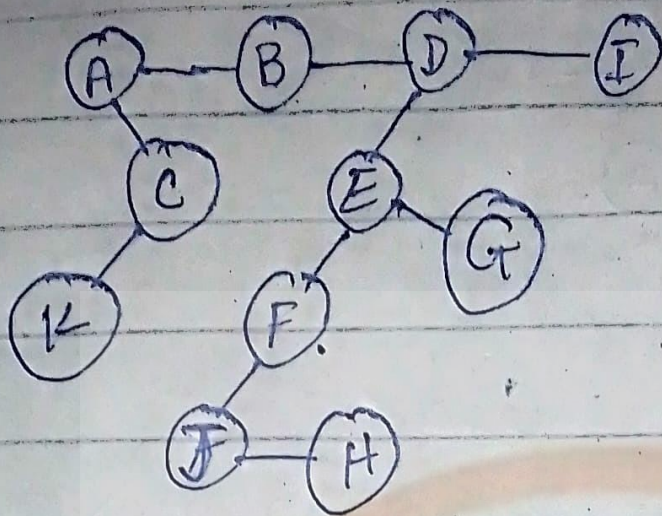
stack: A, C

output: I, H, J, F, G, E, D, B, K, C

stack: A

output: I, H, J, F, G, E, D, B, K, C, A

⇒



DFS-Tree ✓

DFS-Spanning Tree. ✓